Test Point Design

ICT • FCT • JTAG • Boundary scan — testable boards win

Designing for production test from day one

Audience: Engineers preparing for volume production

Why • Types • Placement • Bed-of-nails • JTAG • Cost

Why Test Points

Production test

Bed-of-nails fixture
verifies every board
before shipping.

Bring-up

Probe rails, signals
during debug.

Field service

Diagnostics in deployed
product.

Boundary scan

JTAG-tested
interconnects.

Burn-in

Apply test vectors
during high-temp test.

Without them

Can't ICT —
board fails to ship.

Test Point Types

- ICT test point: 1 mm diameter copper pad (bare) for bed-of-nails pin
- FCT test point: header pin / connector for functional test connection
- JTAG/SWD: 2-10 pin header for boundary scan + debug
- Test pad on a net: ICT only — small Cu pad with no solder mask
- Via test point: stitched-via in pad — pogo pin probes from below
- Through-hole test point: pin sticking up for clip-on probes
- Spring-loaded probe (pogo pin) targets: 1.0-2.0 mm pad diameter

ICT Test Point Specs

- Bare copper round pad
- Diameter: 1.0 mm minimum (1.5 mm typical for reliability)
- Solder mask opening: same diameter as pad (no mask)
- Placement: top side, on the side bed-of-nails accesses
- Pad spacing: ≥ 2 mm between test points (pogo pin spacing constraint)
- Spring force: pogo pins exert ~50-100 g; pad must be on a stiff PCB area
- Avoid placing test points over flex circuits or warped areas
- Mark test points clearly on silkscreen (TP1, TP2, …)

What to Test = Test Point Plan

- Every voltage rail: TP_5V, TP_3V3, TP_1V8, TP_VBAT, TP_VREF, etc.
- Ground: TP_GND × 2-3 around the board (probe attaches close to signals)
- Every reset signal: TP_RST
- Boot/strap pins: TP_BOOT, TP_HOLD
- Critical comm lines: TP_UART_TX/RX, TP_I2C_SDA/SCL, TP_SPI_*
- ADC inputs: TP_ADC0, TP_ADC1, …
- DAC outputs: TP_DAC0, …
- Critical control signals (motor enable, PWM, etc.)

Bed-of-Nails Fixture Design

- Custom fixture made for each board design
- PCB or aluminium plate with vertical pogo pins matching every test point
- Pneumatic press lowers board onto pins
- Pin force: 50-100 g per pin (spring-loaded pogo)
- Cost: ₹50-200k typical for production-grade fixture
- Test time: seconds per board
- Best for: medium-to-high volume production (≥ 1000 pcs)
- For prototype / low-volume: use flying-probe instead (no fixture)

FCT — Functional Test

- FCT = Functional Circuit Test — board powered up + firmware run
- Connects via test header (e.g., 6-pin pogo connector or USB)
- MCU runs production test firmware (separate from product firmware)
- Tests: comms, sensor readings, LED states, calibration constants
- Pass/fail decision per board within seconds
- Test header should expose: power, ground, UART debug, JTAG/SWD
- FCT often combined with ICT — ICT first, then FCT on passing boards

JTAG / Boundary Scan

- JTAG = standard interface for chip-internal test + programming
- 5-pin: TDI, TDO, TCK, TMS, TRST (some skip TRST)
- SWD (ARM): 2-pin Serial Wire Debug — used in Cortex-M MCUs
- Boundary scan: tests interconnections between JTAG-capable chips
- Use for: programming the MCU, debugging firmware, ICT-style net check
- Header: 10-pin (0.05") or 20-pin (0.1") Cortex Debug Connector standard
- Save space: pogo-pin-only test points (no fixed connector after testing)

Placement Best Practices

- Group test points together (one ICT region) for compact fixture
- Pitch: 2.54 mm or 2 mm typical for ICT pads
- Tighter pitch (1.27 mm) possible for premium fixtures
- Don't place test points UNDER large components (BGAs, modules)
- Top side preferred — bed-of-nails contacts top
- Bottom-side test points possible if fixture has top + bottom pin set
- Locate near nets that need testing — minimize trace to test pad
- Don't place near tall components (would damage pogo pins)

Test Coverage Calculation

- Test coverage = % of nets reachable from test points
- ICT typical coverage: 80-95% (some nets inaccessible due to BGA balls)
- Boundary scan coverage: 70-90% (covers JTAG-capable chip pins)
- Combined ICT + boundary scan: 95-99%
- Improve coverage: add test pad on EVERY important net during design
- Don't add test point to every signal pin — would overwhelm the fixture
- Strategic: only add on power, reset, comm, critical signals
- Tool: ASSET InterTech ScanWorks does coverage analysis

Common Test Point Mistakes

- Forgetting test points → can't ICT → can't ship at volume
- Test points on inner layers (not accessible by bed-of-nails)
- Solder mask on test pads → bad pogo-pin contact
- Test points clustered around tall components → fixture can't reach
- Spacing too tight (< 2 mm) → fixture build complexity + cost
- No silkscreen labels → operator confusion during debug
- Reusing test points as connectors → if connector wears, no test possible
- Missing JTAG header → can't program in production

Production Test Workflow

- 1. PCB enters test station, placed on bed-of-nails fixture
- 2. Fixture lowers pogo pins onto test points
- 3. ICT runs: continuity, voltages, resistances — pass/fail in seconds
- 4. If pass: move to FCT station — power up, run test firmware
- 5. FCT verifies functions, calibration, comm — pass/fail in seconds-to-minutes
- 6. If pass: assigned serial number, programmed with production firmware
- 7. Failed boards → repair queue or scrap
- 8. Pass rate metric tracked = First Pass Yield (FPY)

Key Takeaways

Key takeaways from this lecture

Plan test in design

Add test points before layout.

Power + reset always

Plus critical comm + sensor nets.

1 mm bare Cu pads

No soldermask on test pads.

ICT + FCT + JTAG

Combine for full coverage.

Test coverage target ≥ 90%

Strategic placement.

Fixture cost real

₹50-200k — plan early.

Questions & Discussion

Ask anything — concepts, projects, sourcing, career.

References & Further Learning

- IPC-9252 — Requirements for Electrical Testing of Unpopulated Printed Boards
- ASSET InterTech ScanWorks — boundary scan tool
- JTAG Technologies — JTAG fixtures + software
- Acculogic / OK International — fixture vendors in India
- Phil's Lab YouTube — test point design videos
- ARM Cortex Debug Connector specs
- ICT setup videos from Keysight
- Production test best practices — IPC-9252

Thank you • Build something this week.